How-To Library

Informatica

Getting started with data profiling in Enterprise Data Catalog

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Abstract

Use this article to learn how you can use profiling to analyze the content and structure of data sources. You can determine the characteristics of source data in columns, such as value frequency, percentages, and patterns. This article provides information on profiling concepts, architecture components for profiling, profiling process, a use case related to profiling, and steps to get started with profiling.

Supported Versions

- Informatica Enterprise Data Catalog 10.5

Table of Contents

Profiling overview. 2
Key terms in Enterprise Data Catalog profiling. 3
Design-time profiling process 4
Enterprise Data Catalog profiling use case. 4
Checklist for profiling. 5
Step 1. Create resources and enable profiling with data discovery. 6
Step 2. Configure domain connection settings. 7
Step 3. Configure the run-time environment. 8
Step 4. Configure sampling options and other basic profile settings. 8
Step 5. Configure similarity profile and value frequency settings. 10
Step 6. Configure unique key inference settings and run the resource. 11
Step 7. Configure and run the Informatica Similarity Discovery resource. 12
Step 8. View profile results in Enterprise Data Catalog tool. 13
Run-time profiling process. 18

Profiling overview

A profile is a repository object that finds and analyzes data across data sources in the enterprise. Running a profile on any data source in the enterprise gives you a good understanding of the strengths and weaknesses of its data and metadata.

You can use profiling to find the content, quality, and structure of data sources of an application, schema, or enterprise. The benefit of data profiling is that you can identify hidden data problems that put data projects at risk. The data discovery process identifies profiling metadata, such as null values, distinct values, inferred data types, unique keys, and data domains in the resource.

You can use data profiling to perform the following types of data analysis:

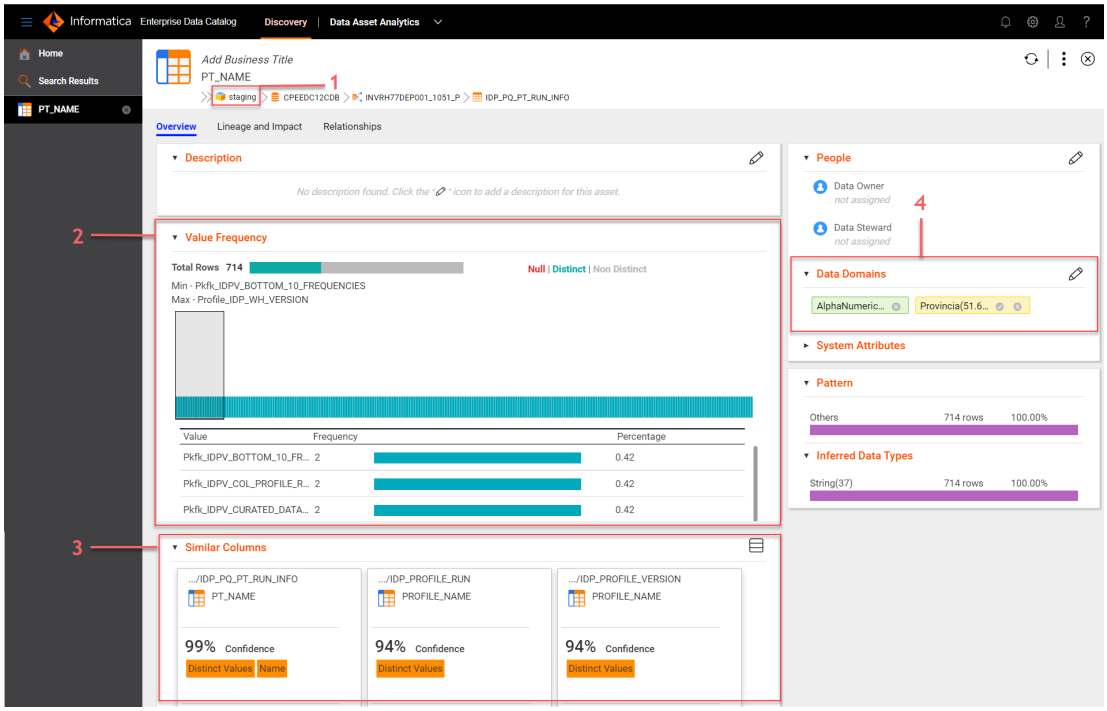
- Column profiling. You can discover the number of unique values, null values, and data patterns in a column.
- View similar columns. You can view similar columns based on column names, column data patterns, and unique values for the data assets.
- View value frequencies. You can enable value frequency along with column data similarity to compute the frequency of values in a data source.

- View inferred or accepted data domains. You can view inferred or accepted data domains for the asset from the profile results or from similar columns.
- Curate data domains. You can accept or reject inferred data domains.

Key terms in Enterprise Data Catalog profiling

You can discover, explore, and relate different types of metadata from disparate sources using the Enterprise Data Catalog tool. The key terms in Enterprise Data Catalog profiling include resource, value frequency, column similarity, and data domain.

You can view the profiling results in the Enterprise Data Catalog tool as shown in the following image:



1. Resource

A resource is a catalog object that represents an external data source or metadata repository from where scanners extract metadata. Each resource can extract both source metadata and profile metadata from the external data sources.

2. Value frequency

Value frequence determines how many times a value appears in the same column. You can identify the frequency of values after you enable column similarity for a resource. Based on your business requirement, you can use the value frequency to analyze data in a resource.

3. Column similarity

As a data analyst or data architect, you can scan your enterprise data to find similar columns. When the scanner runs based on a schedule, it uses the inferred similar columns to propagate smart data domains to other similar columns. This process allows you to search and discover assets of interest faster in the catalog.

4. Data domain

A data domain is a predefined or user-defined model repository object that enables you to discover the functional meaning of column data or column names in a data source. Examples of data domains include Social Security number, account status, IP address, and UPC code.

Design-time profiling process

The profiling process includes data preparation, staging of data, discovery, and ingestion of data in the catalog. By default, all profiles run in the native run-time environment. You can use native data sources to create and run profiles in the native environment.

The design-time profiling process includes the following steps:

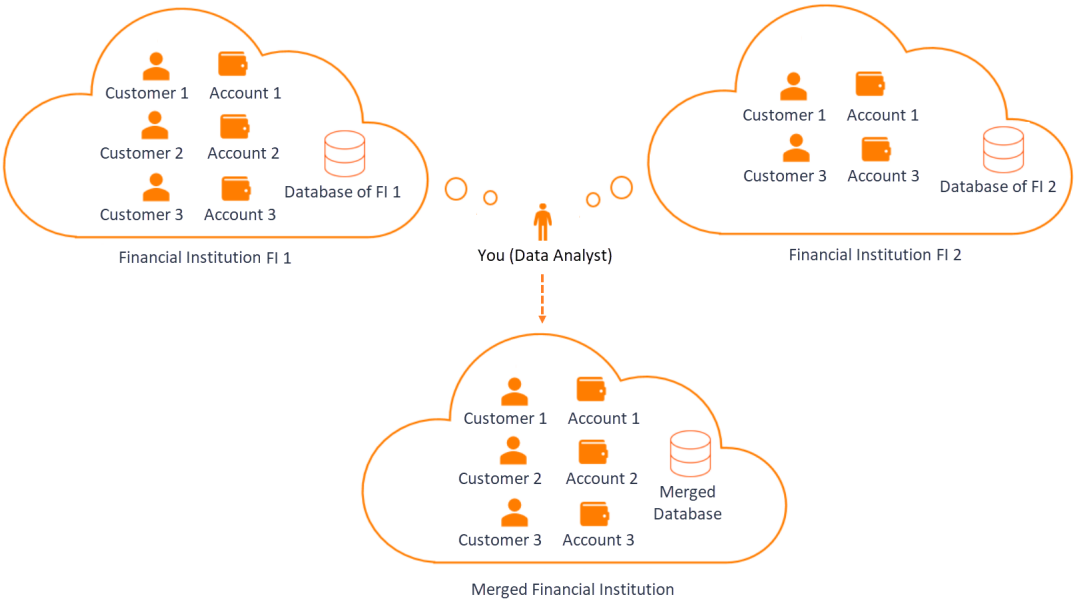
1. Enable data discovery for the resource in the Catalog Administrator tool to discover profiling-related metadata and unique keys.
2. Configure the sampling options, column similarity settings, value frequency settings, and unique key inference settings in the Catalog Administrator tool.
3. Run the resource.
4. Review and validate the profiling results in the Enterprise Data Catalog tool for columns or fields in the resource that support profiling.
5. Curate an inferred data domain in the Enterprise Data Catalog tool based on your requirements. You can accept or reject the inferred data domains.

Enterprise Data Catalog profiling use case

You are a data analyst at a financial institution that has branches and franchises across North America.

The institution recently acquired another financial institution. Some customers hold accounts in both the financial institutions. You and your team are asked to integrate all the customer details in a single database.

The following image shows the tasks you need to perform as a data analyst:



You and your team need to perform the following tasks to extract the required information:

Task	Resolution
Identify data sources when different database systems are used by the financial institution and the acquired institution.	You can identify the data sources that need to be scanned to find the required customers that match the eligibility criteria. You can add these data sources as resources in the Catalog Administrator tool to extract metadata from these resources.
Enable data discovery and similarity profiling when there is lack of consistency and context in the column names that makes it difficult to find and analyze source columns with similar data.	You can perform the following steps: - In the Catalog Administrator tool, you can enable data discovery and similarity profiling for the selected resource. The profiling scanner and Informatica Similarity Discovery resource identifies the similar columns in the resource based on column names, column data patterns, and unique values. - In the Enterprise Data Catalog tool, you can view columns that contain similar data across all the data sources, identify data that can be joined, and duplicate data that can be removed. [1]

1. For example, you find out from an existing bank report that both the organizations store the Social Security number (SSN) on all records that have customer information. Based on this information, you infer that if the tables contain a column with SSN information then the customer details might be present in the tables. To verify, you search for an SSN column in the catalog, the Catalog tool lists the searched column along with other columns from all the data sources that are similar to the searched column. Based on the search results, you and your team can identify data that can be joined and remove duplicate data.

Checklist for profiling

Before you configure profiling for a resource, make sure that the following prerequisites are met:

Checklist	Prerequisite
❑	Verify that the Model Repository Service is configured in the Informatica domain.
❑	Create connections to the profiling warehouse, staging database, and source data.
❑	Integrate the Informatica domain with the Hadoop environment if you want to run profiles on the Blaze or Spark engines.
❑	Add all the environment variables in the <source_filename> source file. Run the following commands to set up the INFA_HOME environment variable for the database client in the domain: - `/INFA_HOME/tomcat/bin/infaservice.sh shutdown` - `source <source_filename>` - `/INFA_HOME/tomcat/bin/infaservice.sh startup`
❑	Verify that you have the **Data Privileges: View Data** privilege before you configure the value frequency settings for a resource.
❑	Assign **Metadata and Data Read** or **All Permissions** to the resource and assign **Read** permission to the DataDomain resource.

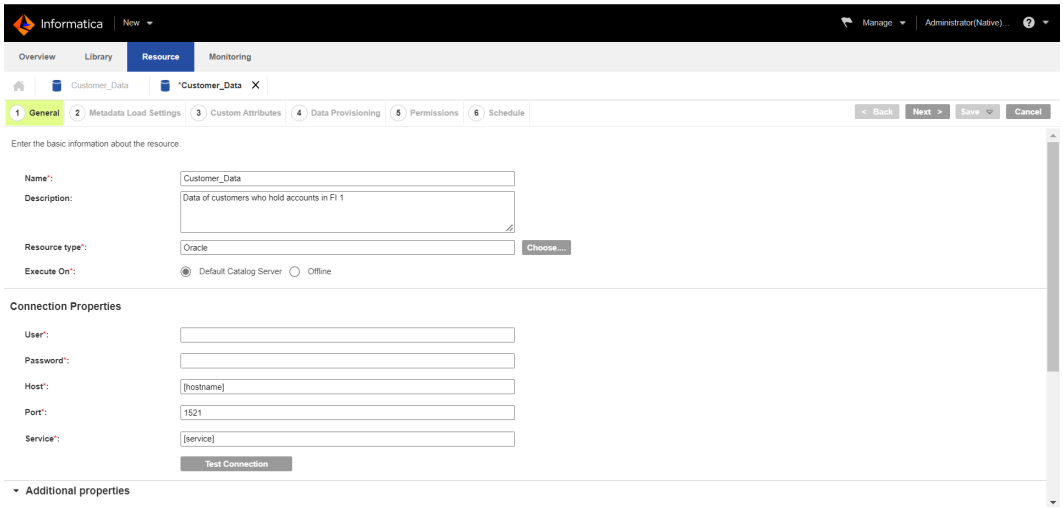
Step 1. Create resources and enable profiling with data discovery

Create resources in the Catalog Administrator tool and enable profiling with data discovery to discover profiling-related metadata and unique keys from the different data sources. You can add the different database systems used by the financial institution and the acquired institution as resources to extract metadata.

First, create a resource **Customer_Data** in the Catalog Administrator tool for the main financial institution and enable profiling with data discovery.

1. Log in to the Catalog Administrator tool.
2. Click **New** > **Resource**.
3. Enter the resource name as **Customer_Data** and specify an optional description for the resource.
4. Click **Choose** to open the **Select Resource Type** dialog box.
5. Select a resource type, and click **Select**.
6. Specify the connection properties for the resource.

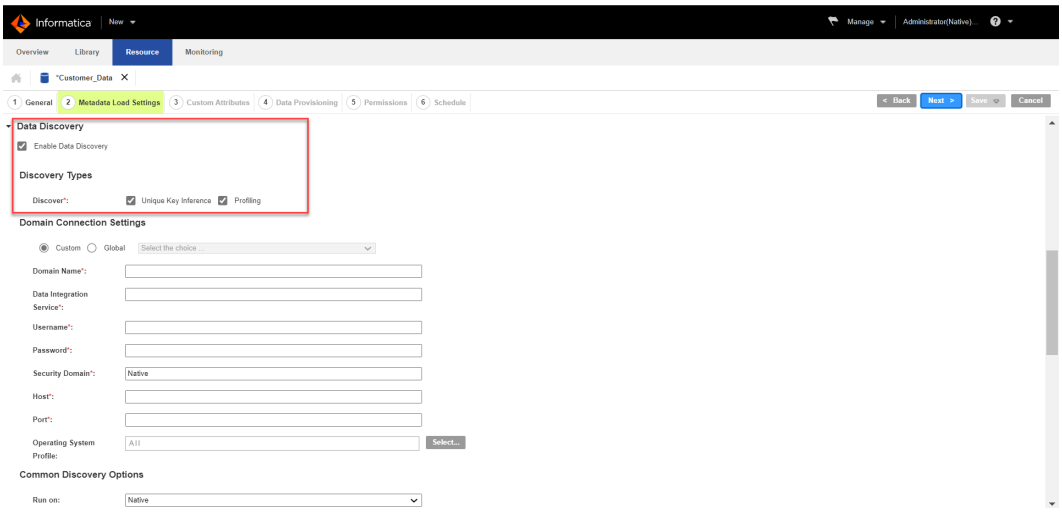
   The following image shows the general configurations of the **Customer_Data** resource:



7. Click **Next** to move to the **Metadata Load Settings** page.
8. Select the **Enable Data Discovery** option to perform data discovery on the resource.

9. Select **Profiling** and **Unique Key Inference** options in the **Discovery Types** section to enable profiling for the resource and to uniquely identify rows in the data source.

   The following image shows that unique key inference and profiling is enabled for the **Customer_Data** resource:

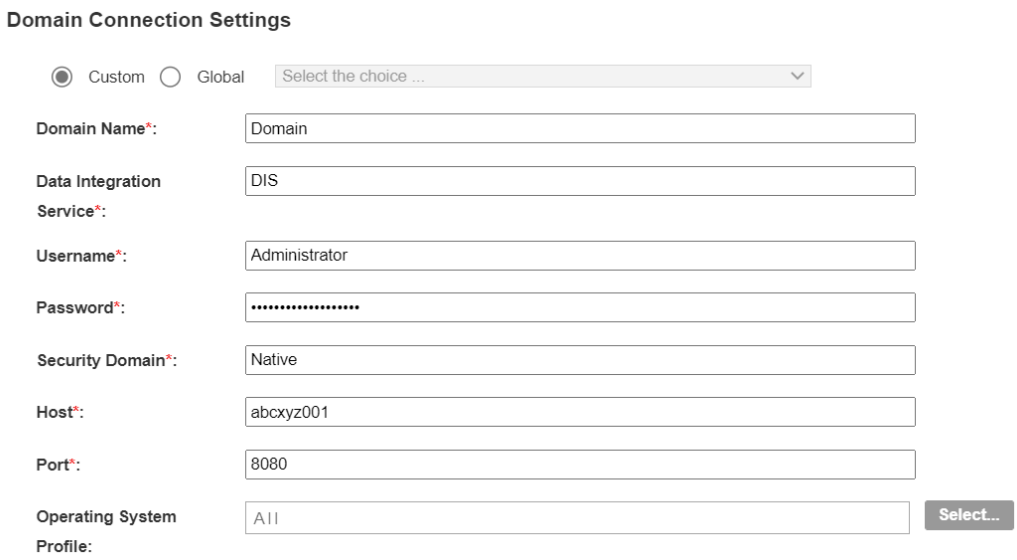


Step 2. Configure domain connection settings

Configure the domain connection settings for the Data Integration Service.

1. In the **Domain Connection Settings**, select **Custom** and enter the domain name.
2. Enter the name of the Data Integration Service.
3. Enter your user name and password.
4. Enter the name of the security domain. By default, **Native** is the security domain name.
5. Enter the domain host name and port number.

   The following image shows the domain connection settings configured for the **Customer_Data** resource:



Step 3. Configure the run-time environment

Configure the run-time environment for the **Customer_Data** resource to run the profile.

1. In the **Common Discovery Options**, select **Native** as the run-time environment. This runs the profile on the same machine where the Data Integration Service runs.

   Note:

   - You can also run the profile in the Hadoop environment on the Blaze or Spark engine.
   - You can also run the profile in the Databricks environment on the Databricks Spark engine in the Databricks cluster.

   The **Common Discovery Options** section of the **Metadata Load Settings** page appear as shown in the following image:

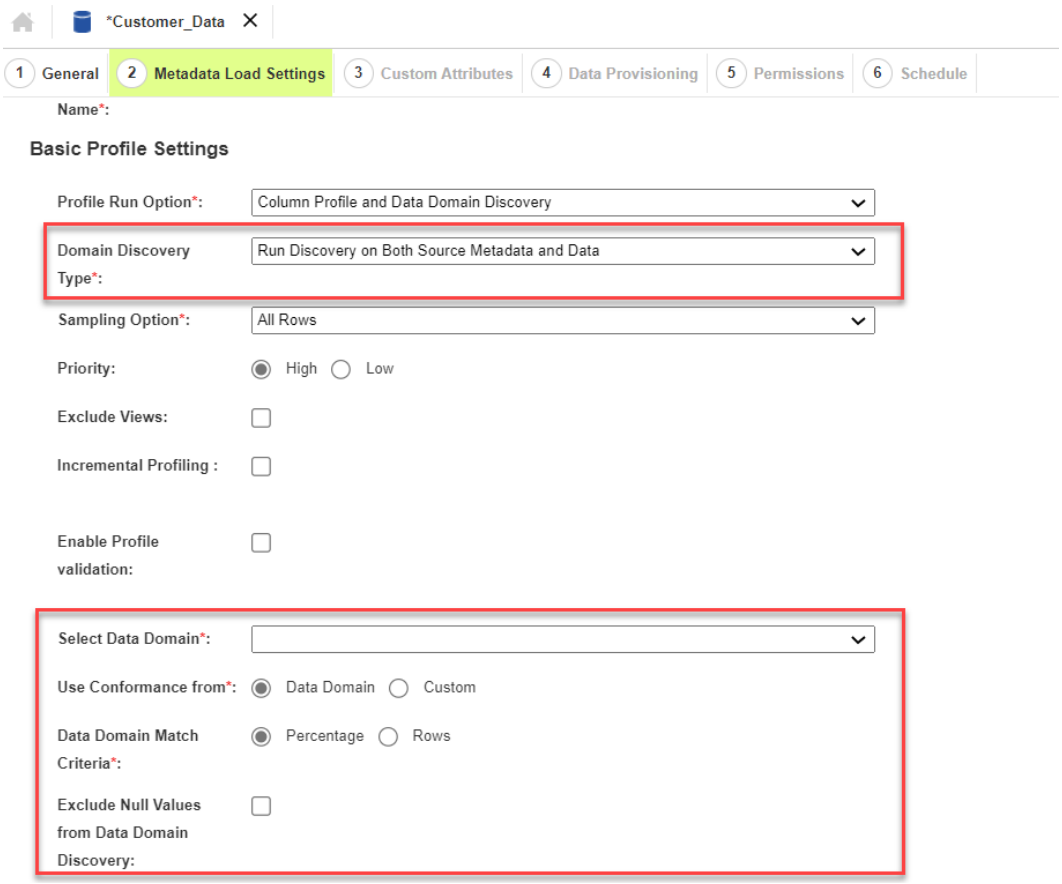
   

2. Click **Select** and select the **Source Connection Name** from the **Source Connection Name** window.

Step 4. Configure sampling options and other basic profile settings

Configure sampling options for the **Customer_Data** resource to determine the number of rows in the data source to run the profile job on.

1. In the **Basic Profile Settings** section, select **Column Profile and Data Domain Discovery** as the option for the profiling scanner to run the profile job on the resource. This identifies the number of null values, distinct values, non-distinct values, and infers data patterns, data types, and data domains in the resource.
2. When you choose **Data Domain Discovery** or **Column Profile and Data Domain Discovery** as the profile run option, the **Basic Profile Settings** section refreshes with more fields as highlighted in the following image :



3. Select **Run Discovery on both Source Metadata and Data** as the data domain discovery type for the profiling scanner to infer data domains. This runs the data domain discovery on both the source data and source metadata.

   Note: You can choose only the **Run Discovery on Source Data** option when you run data domain discovery on unstructured data sources.

4. Select **All Rows** as the sampling option. This runs the profile on all the rows in the data source.
5. Select **High** as the priority for the profiling scanner to prioritize the resource run.

   Note: The profiling scanner runs the resources with **High** priority first, and then runs the resources with **Low** priority.

6. Select **All Data Domains** from the **Select Data Domain** drop-down list. This discovers all the data domains in the resource.
7. Select **Data Domain** from the **Use Conformance from** drop-down list. This uses the predefined conformance values that you configured for the data domains.

   When you create a data domain, you can configure the minimum percentage of source rows and minimum number of source rows as the conformance criteria for data domain match. These values are predefined conformance values.

8. Choose **Percentage** as the conformance criteria for the data domain match. This computes the conformance criteria as the ratio of the number of matching rows divided by the total number of rows.

Step 5. Configure similarity profile and value frequency settings

Configure similarity profile and value frequency settings for the **Customer_Data** resource to identify the frequency of values in the data sources.

1. In the **Similarity Profile Data Preparation and Value Frequency Settings** section, choose **Yes** as the **Run Similarity Profile**.

   The profiling scanner scans the data source and prepares data to perform the following tasks:

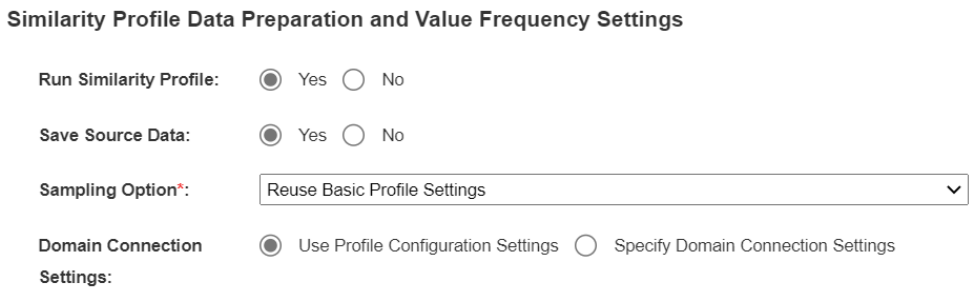
   - Discover similar columns. The algorithm discovers similar columns in the resource based on column names, column patterns, and unique values.
   - Identify business terms. The algorithm identifies and recommends the business terms for a column based on the accepted data domains and similar columns.

2. Choose **Yes** in the **Save Source Data** option. This persists the computed information about similar columns, column patterns, and unique values in the resource in PostgreSQL database.

3. Select **Reuse Basic Profile Settings** as the sampling option. This uses the sampling option specified in the **Basic Profile Settings** section.

4. Choose **Use Profile Configuration Settings** as the domain connection settings. Enterprise Data Catalog uses the Data Integration Service specified in the **Domain Connection Settings** section to identify similar columns in the data sources.

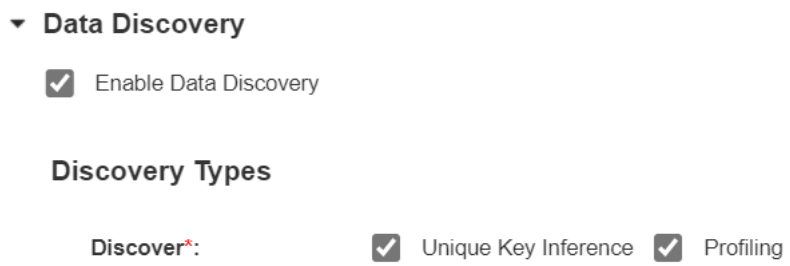
   The following image shows the **Similarity Profile Data Preparation and Value Frequency Settings** section configured for the **Customer_Data** resource:



Step 6. Configure unique key inference settings and run the resource

Configure the unique key inference settings for the **Customer_Data** resource to uniquely identify rows in the data source.

1. Select the **Unique Key Inference** option in the **Discovery Types** section to generate unique key candidate from selected columns for the resource as shown in the following image:

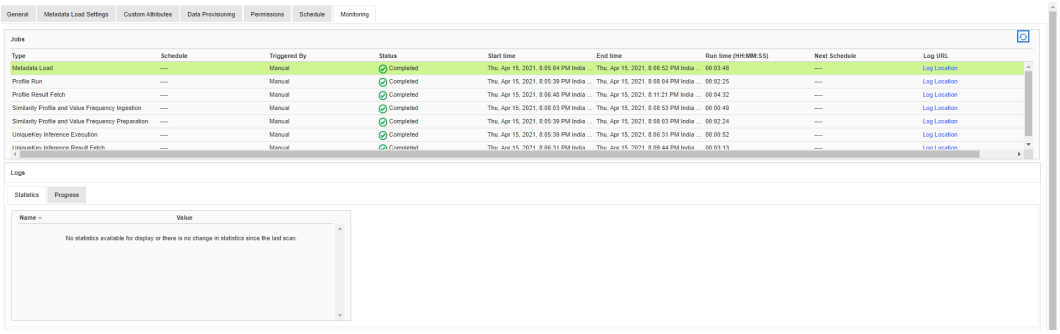


When you select the **Unique Key Inference** option, the **Unique Key Inference Settings** section appear as shown in the following image:



2. In the **Unique Key Inference Settings** section, select **Yes** to run the unique key inference. The profile scanner scans and infers unique keys from the data source.
3. Set a threshold for null values in the unique key inference. You can enter a value between 0 and 1.
4. Select **Yes** to skip the table with the documented or accepted unique keys.
5. Select **All Rows** as the unique key sampling option. This runs the unique key inference on all the rows in the data object.

6. Click **Save and Run**. The **Customer_Data** resource details appear on a new page with the **Monitoring** tab enabled as shown in the following image:



Note: After you create the resource, assign **Metadata and Data Read** or **All Permissions** to the resource and assign **Read** permission to the DataDomain resource.

Similarly, create another resource **Customer_Data_FI2** in the Catalog Administrator tool for the acquired financial institution and enable profiling with data discovery.

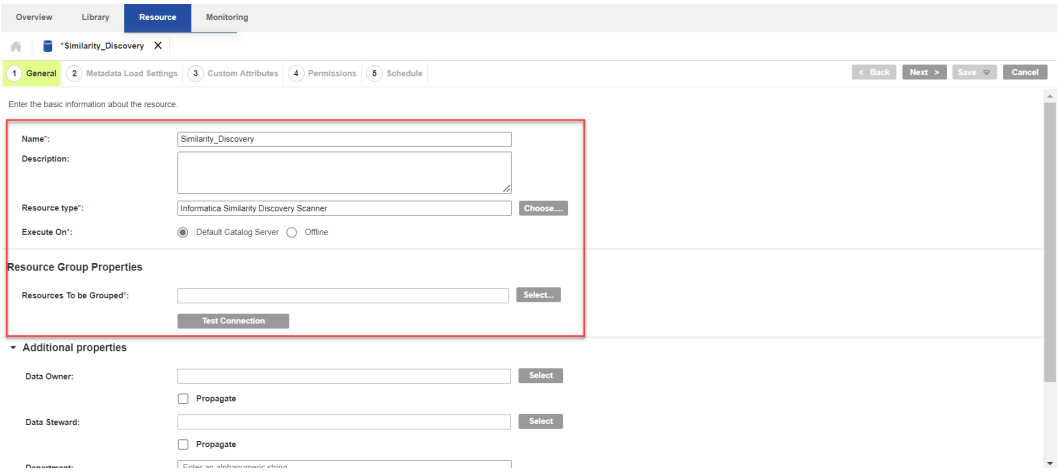
For more information on creating resources, see the *Informatica Enterprise Data Catalog Administrator Guide*.

Step 7. Configure and run the Informatica Similarity Discovery resource

Configure and run the **Informatica Similarity Discovery** resource to discover similar columns across the data sources of the main and the acquired financial institutions.

1. In the Catalog Administrator tool, click **New** > **Resource**.
2. Enter the resource name as **Similarity_Discovery** and specify an optional description for the resource.
3. Click **Choose** to open the **Select Resource Type** dialog box.
4. Select the resource type as **Informatica Similarity Discovery Scanner**.
5. In the **Resources To be Grouped**, click **Select** to open the **Select Resources To Be Grouped** dialog box.
6. Select the previously created **Customer_Data** and **Customer_Data_FI2** resources and click **Select**.

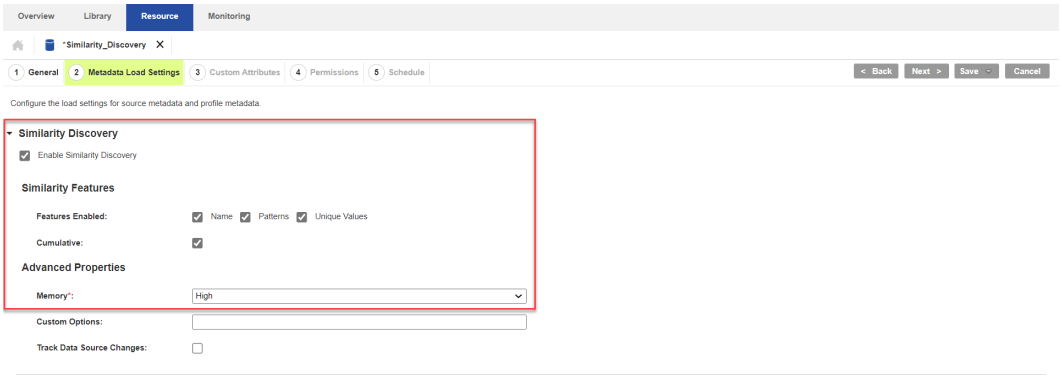
   The following image shows the general configurations of the **Similarity_Discovery** resource:



7. Click **Next** to move to the **Metadata Load Settings** page.
8. Select the **Enable Similarity Discovery** option to run similarity discovery on the selected resources.

9. In the **Features Enabled** section, select **Name**, **Patterns**, and **Unique Values** to discover similar columns based on column name, column data pattern, and unique values in the resource.
10. In the **Advanced Properties** section, select the **Memory** required to run the resource as **High**.

    The following image shows the metadata configurations of the **Similarity_Discovery** resource:



11. Click **Save and Run**. The **Similarity_Discovery** resource details appear on a new page with the **Monitoring** tab enabled.

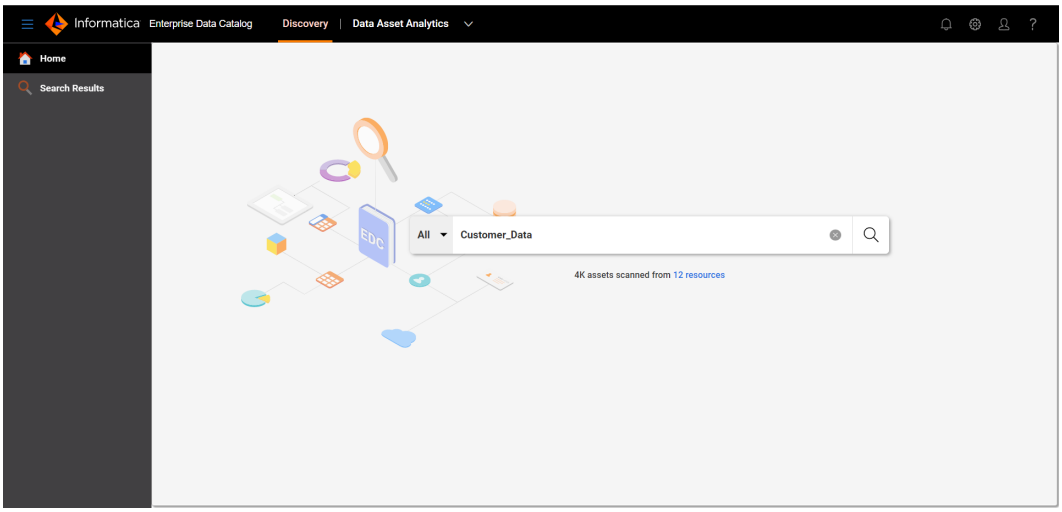
    For more information on the **Informatica Similarity Discovery** resource, see the *Informatica Similarity Discovery Resource* section in the *Informatica Enterprise Data Catalog Scanner Configuration Guide*.

Step 8. View profile results in Enterprise Data Catalog tool

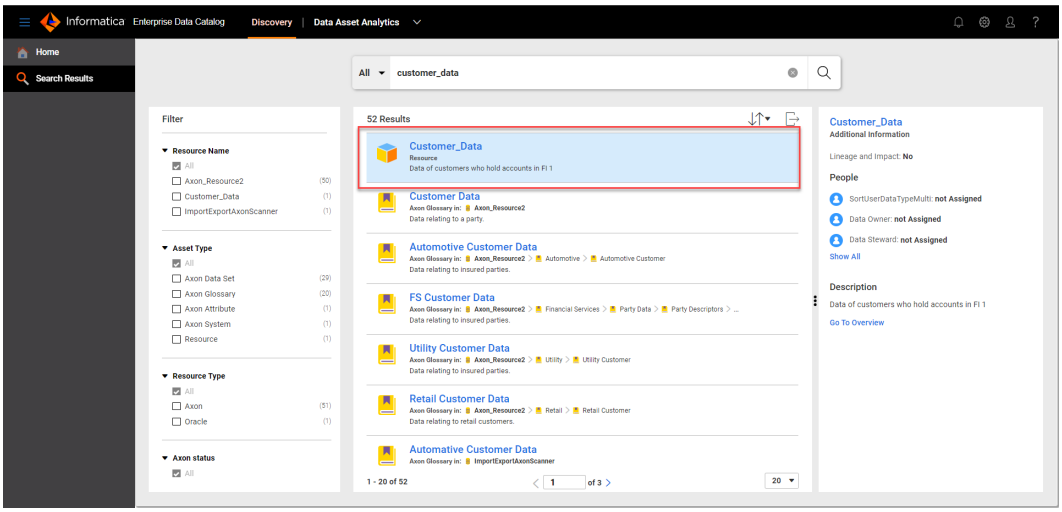
After you configure the resource with profiling, you can view the profiling results in the Enterprise Data Catalog tool and view columns that contain similar data across all the data sources, identify data that can be joined, and duplicate data that can be removed.

1. Log in to the Enterprise Data Catalog tool.

   The Enterprise Data Catalog search page appears.
2. Type **Customer_Data** in the search box.



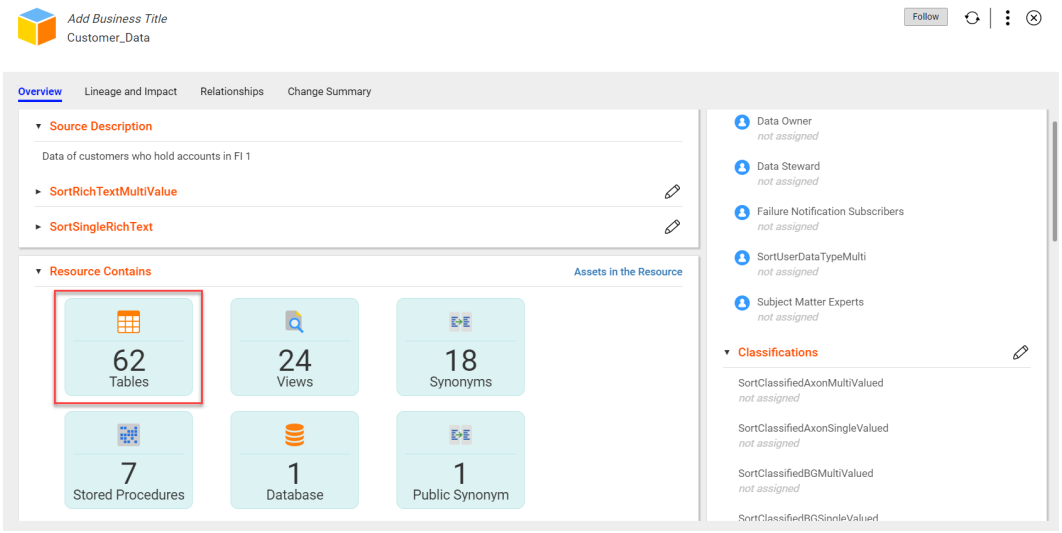
The **Customer_Data** resource appears on the **Search Results** page as shown in the following image:



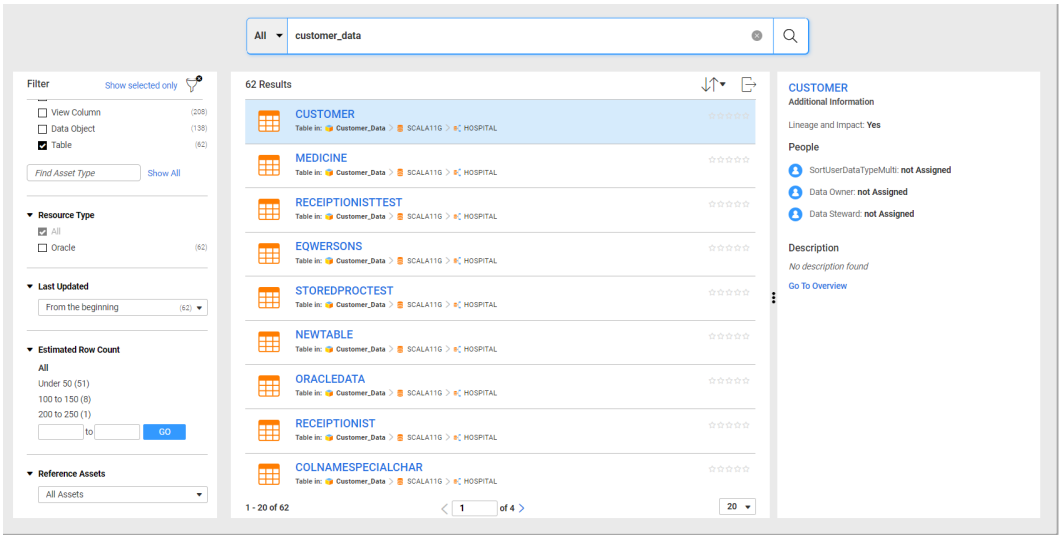
3. From the **Search Results** page, open the **Customer_Data** resource.

   The resource opens in the **Overview** tab.

4. In the **Overview** tab of the resource, click **Tables** from the **Resource Contains** section as shown in the following image:



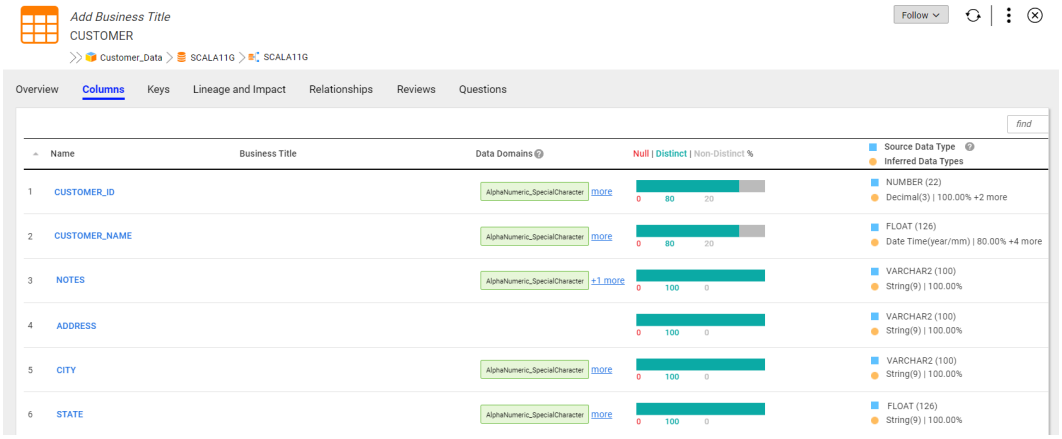
The tabular assets appear on the **Search Results** page as shown in the following image:



5. From the **Search Results** page, open the **CUSTOMER** table.

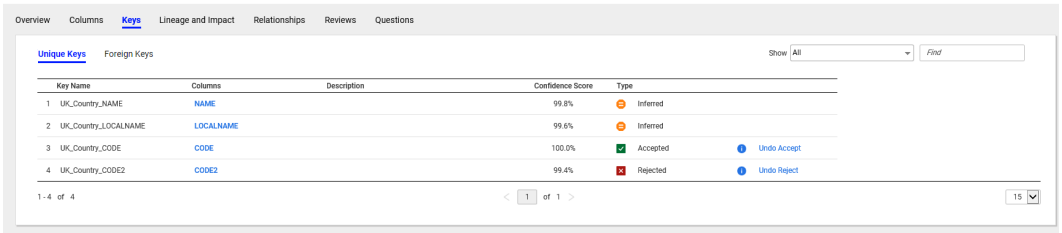
   The **CUSTOMER** table opens in the **Overview** tab.

6. From the **Overview** tab of the **Customer** table, open the **Columns** tab to view profiling results as shown in the following image:



7. Click on the **Keys** tab to view details of the unique keys corresponding to the table.

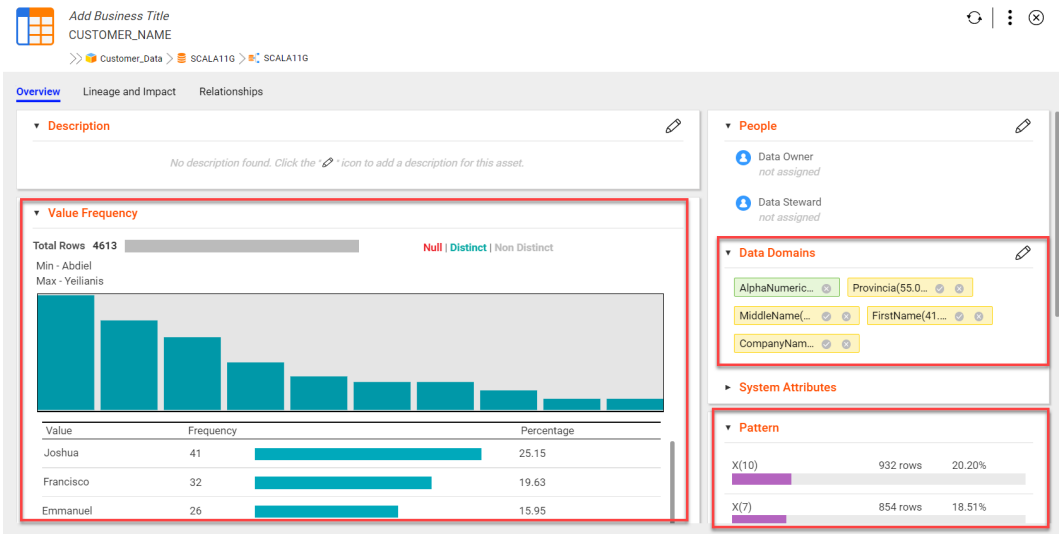
   The following image displays the unique keys in the **CUSTOMER** table:



You can curate the unique key results. Hover the mouse over the key name to accept or reject the columns inferred as the unique key. The inferred type appears in orange color, accepted type appears in green color, and rejected type appears in red color. You can click **Undo Accept** or **Undo Reject** to reset the unique key results to the inferred state.

8. To view value frequency details, inferred data domains, and data pattern, click the column name from the **Columns** tab.

   The following image displays value frequencies, data domains, and pattern information of the **CUSTOMER_NAME** column in the **CUSTOMER** table:



9. You can accept or reject the inferred data domains. The inferred data domains appear in orange boxes with the inference percentage. For example, 

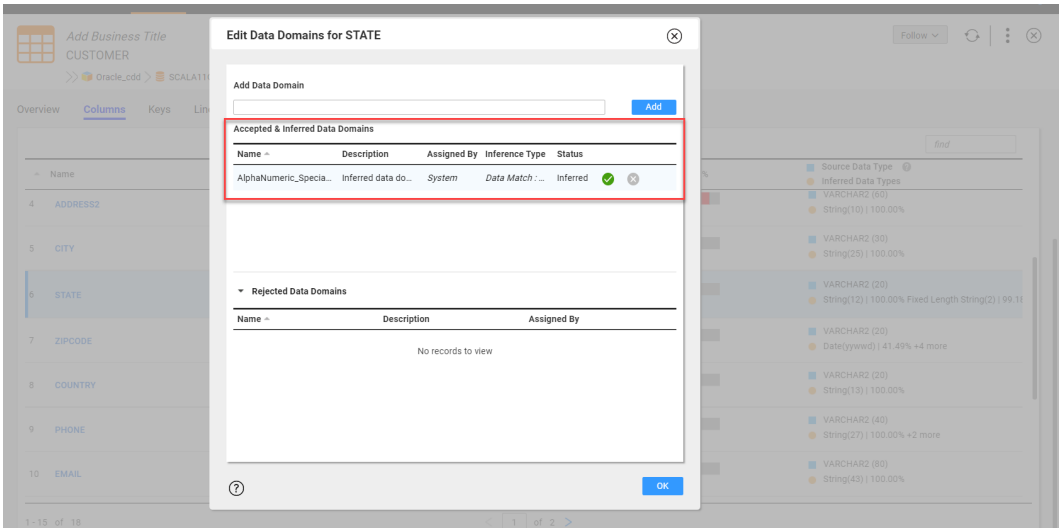
. The data domains that you accept are assigned to the asset. The assigned data domains appear in green boxes. You can reject any assigned data domain, if required.

   You can perform the following tasks on an inferred data domain:

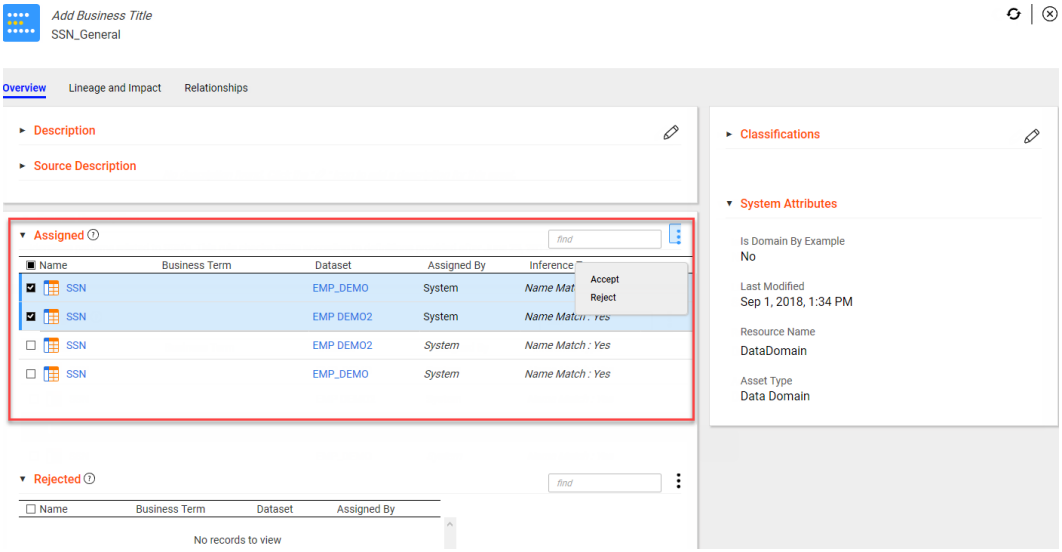
   - Click the icon to accept the data domain. After you accept the data domain, the box turns green. The data domains that are auto-accepted appear in a green box. For example, 

.
   - Click the icon to reject the data domain. After you reject a data domain, the data domain no longer appears in the **Data Domain** section for the asset.

You can also curate the inferred data domains at the table level. The following image displays the curation of data domains from the **CUSTOMER** table:

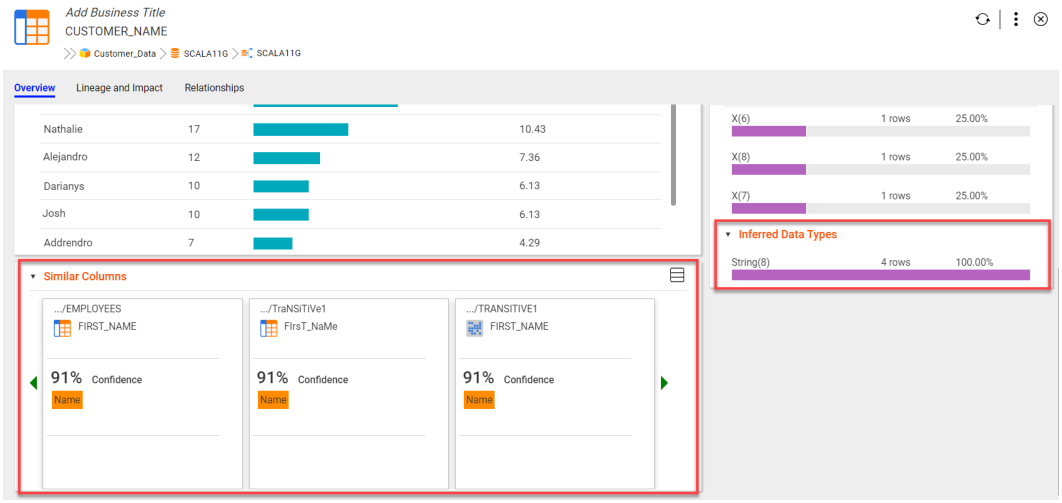


You can also curate the inferred data domains in bulk at the data domain level. The following image displays the bulk curation of the **SSN_General** data domain:



You can also curate the data domains using export functionality at the resource, table, column, and data domain level. You can export the assets to the comma-separated values (CSV) file and import the CSV file back into Enterprise Data Catalog. The exported CSV file has inferred, accepted, and rejected data domains column headings. You can edit the columns to accept, reject, or delete the assets.

You can view the columns similar to a specific column and the inferred data types in the **Columns** tab. The following image displays similar columns and inferred data types of the **CUSTOMER_NAME** column in the **CUSTOMER** table:



For more information on viewing the profiling results, see the *Informatica Enterprise Data Catalog User Guide*.

Using the above profiling results, you can perform the following:

- View columns that contain similar customer data across all the data sources of both financial institutions.
- Associate inferred data domains to discover the functional meaning of column data.
- Identify customer data that can be joined.
- Remove duplicate data from the data source.

If you find similar data that you want to join into a single customer record or delete duplicate records, you can either rectify the data directly in the data source or use Informatica Data Integration. To learn more about joining similar data from different data sources, or removal of duplicate data from a data source, see the *Informatica Data Engineering Integration User Guide*.

Run-time profiling process

When you run a profile, the Data Integration Service runs the profiles and writes the profile results to the profiling warehouse.

You can achieve the following data objectives during run time with profiling:

1. The Catalog Service gets the profile definition from the Model repository and establishes a connection to the data source.
2. The Catalog Service invokes the Profiling module in the Data Integration Service.
3. The Profiling module processes the profile job and submits the job to the Data Integration Service.
4. The Data Integration Service generates and writes the profiling results to the profiling warehouse.

Author

Sabha S Nasmin N S